

Physics, 2018 EC, entrance exam, version 3

Grade 12 · Physics · 2018 E.C.

Time allowed: 120 minutes**Questions:** 48**Total marks:** 48

Name _____ Grade _____ ID No. _____ Date _____

1. When an object moves along a curved path between two points, the distance traveled is usually greater than the displacement because: (1)
- A. A. Distance measures the shortest path.
 - B. B. Displacement measures the total path length.
 - C. C. Distance always equals displacement.
 - D. D. Distance measures the entire path traveled while displacement measures the shortest straight line.
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2. During a journey, the velocity of a moving object may change continuously due to acceleration or deceleration. Which of the following quantities helps describe the velocity of the object at a specific moment during such motion? (1)
- A. A. Average velocity
 - B. B. Instantaneous velocity
 - C. C. Total displacement
 - D. D. Average speed
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3. Which of the following statements correctly defines uniform motion? (1)
- A. A. Motion in which an object covers equal distances in equal intervals of time
 - B. B. Motion in which an object changes its speed continuously
 - C. C. Motion in which the object stops after some time
 - D. D. Motion in which the object changes direction frequently
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4. Which of the following examples represents a force acting without direct contact between objects? (1)
- A. A. A person pushing a table
 - B. B. A magnet attracting small iron pieces
 - C. C. A hammer hitting a nail
 - D. D. A hand pulling a rope
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5. A student applies a force to push a heavy wooden box across the floor, but the box moves very slowly compared to a lighter box pushed with the same force. Which of the following explanations best accounts for this observation? (1)
- A. A. Acceleration decreases when the mass of the object increases for the same force
 - B. B. Heavier objects always move faster
 - C. C. Force has no effect on heavy objects
 - D. D. Distance traveled determines acceleration
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6. A boy is rolling a 0.5 kg ball on the frictionless floor with the speed of 20 m/s. The ball gets deflected by an obstacle on the way. After deflection it moves 5% of its initial kinetic energy. What is the speed of the ball now? (1)
- A. A. 19 m/s
 - B. B. 14.4 m/s
 - C. C. 4.47 m/s
 - D. D. 1 m/s

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7. A ramp is used to push a heavy box into the back of a truck. Instead of lifting the box vertically, the worker pushes it along the inclined surface. The ramp makes the work easier mainly because it: (1)
- A. A. Reduces the distance moved by the box
 - B. B. Changes the direction of motion only
 - C. C. Increases the mass of the object
 - D. D. Reduces the force needed by increasing the distance over which the force is applied
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8. A simple wheel and axle is used to lift a 30 N bucket of water out of a well. The radii of the wheel and axle are 20 cm and 5 cm, respectively. If the effort required is 10 N, what is the efficiency of the machine? (1)
- A. A. 75%
 - B. B. 60%
 - C. C. 80%
 - D. D. 90%
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9. A person uses a long lever to lift a heavy rock. The effort applied at the end of the lever moves through a relatively large distance while the rock moves a much shorter distance. Which of the following best describes the primary purpose of the lever in this case? (1)
- A. A. It acts as a speed (distance) multiplier, allowing the effort to move a large distance
 - B. B. It acts as a force multiplier, allowing a small effort to lift a heavy load
 - C. C. It reduces the work done by eliminating gravity
 - D. D. It acts as a direction changer to move the rock horizontally
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10. Two types of waves differ in how particles of the medium move relative to the direction of wave travel: in one type, particles vibrate parallel to the direction of propagation, while in the other, they vibrate perpendicular to it. Which pair correctly identifies these two wave types? (1)
- A. A. Transverse (parallel) and Longitudinal (perpendicular)
 - B. B. Longitudinal (parallel) and Transverse (perpendicular)
 - C. C. Both are parallel
 - D. D. Both are perpendicular
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11. The speed of sound in air at 0°C is approximately 331 m/s. If the temperature of air rises to 20°C, what is the speed of sound at this temperature? (1)
- A. A. 343 m/s
 - B. B. 331 m/s
 - C. C. 341 m/s
 - D. D. 351 m/s
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12. A stone is thrown vertically upward from the ground with an initial velocity of $u = 30 \text{ m/s}$. Assume the acceleration due to gravity is $g = 10 \text{ m/s}^2$ and neglect air resistance. What is the maximum height reached by the stone? (1)
- A. A. 30 m
 - B. B. 40 m
 - C. C. 45 m
 - D. D. 60 m
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13. The length of a wire is increased by 1 mm on the application of a certain load. In a wire of the same material but of twice the length and half the radius, the same force will produce an elongation of (1)
- A. A. 0.5 mm
 - B. B. 2 mm
 - C. C. 4 mm
 - D. D. 8 mm
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14. A rigid ball of weight 100 N is suspended with the help of a string. The ball is pulled by a horizontal force 'F' such that the string makes an angle of 30° with the vertical. What is the magnitude of force 'F'? (1)
- A. A. 47.33 N
 - B. B. 50.82 N
 - C. C. 57.73 N
 - D. D. 67.82 N

15. The electrostatic force on a small sphere of charge $0.4\ \mu\text{C}$ due to another small sphere of charge $-0.8\ \mu\text{C}$ in air is $0.2\ \text{N}$. What is the distance between the two spheres? (1)

A. A. $0.08\ \text{m}$ B. B. $0.12\ \text{m}$
C. C. $0.20\ \text{m}$ D. D. $0.36\ \text{m}$

16. The current through a 10-ohm resistor is 1.2 amperes. What is the potential difference across the resistor? (1)

A. A. $12\ \text{V}$ B. B. $14\ \text{V}$
C. C. $120\ \text{V}$ D. D. $8.3\ \text{V}$

17. A straight long wire carries an electric current to the right. The current is placed in a uniform magnetic field directed into the page. What is the direction of the magnetic force on the current? (1)

A. A. Left. B. B. Right.
C. C. To the bottom of the page. D. D. To the top of the page.

18. The refractive index of a medium is 1.5 . If the speed of light in vacuum is $3\times 10^8\ \text{m/s}$, what is the speed of light in the medium? (1)

A. A. $1\times 10^8\ \text{m/s}$ B. B. $2\times 10^8\ \text{m/s}$
C. C. $2.5\times 10^8\ \text{m/s}$ D. D. $4.5\times 10^8\ \text{m/s}$

19. Which of the following sequences correctly arranges the electromagnetic waves in order of increasing frequency? (1)

A. A. Radio $\rightarrow$ Microwaves $\rightarrow$ Infrared $\rightarrow$ Visible $\rightarrow$ Ultraviolet $\rightarrow$ Gamma rays
B. B. Gamma rays $\rightarrow$ Ultraviolet $\rightarrow$ Visible $\rightarrow$ Infrared $\rightarrow$ Microwaves $\rightarrow$ Radio
C. C. Infrared $\rightarrow$ Radio $\rightarrow$ Microwaves $\rightarrow$ Ultraviolet $\rightarrow$ Visible $\rightarrow$ Gamma rays
D. D. Radio $\rightarrow$ Ultraviolet $\rightarrow$ Microwaves $\rightarrow$ Infrared $\rightarrow$ Visible $\rightarrow$ Gamma rays

20. When a scientist observes an unexpected result in an experiment, the next scientific step is to: (1)

A. A. revise the hypothesis or test alternative explanations.
B. B. conclude the theory is wrong without further testing.
C. C. report it as a failure.
D. D. ignore the result as an anomaly.

21. Two people pull a cart with forces of $30\ \text{N}$ and $40\ \text{N}$ at a 90° angle to each other. What is the type and magnitude of the resultant vector? (1)

A. A. $50\ \text{N}$, coplanar vector B. B. $70\ \text{N}$, concurrent vector
C. C. $10\ \text{N}$, zero vector D. D. $100\ \text{N}$, collinear vector

22. A particle is moving in a straight line with constant acceleration $0.2\ \text{m/s}^2$. After it has moved a total of $20\ \text{m}$, its speed is $8\ \text{m/s}$. What was the initial speed of the particle? (1)

A. A. $5\ \text{m/s}$ B. B. $6\ \text{m/s}$
C. C. $7.5\ \text{m/s}$ D. D. $8.5\ \text{m/s}$

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- 23.** A small ball attached to the end of a string experiences a uniform circular motion on a horizontal frictionless table. In order to keep the ball moving in a circle the string must apply force on the ball toward the center of the circle. Which of the following refers to a reaction force? (1)
- A. A. The force of gravity pulling down
 - B. B. The normal force pushing up
 - C. C. The force applied by the ball on the table
 - D. D. The force applied by the ball on the string
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- 24.** A 2.0 kg block is released from rest at the top of a frictionless incline that is 5.0 m above the ground. The block slides down the incline and reaches the bottom. Which of the following is the correct speed of the block at the bottom? (1)
- A. A. 7.0 m/s
 - B. B. 10 m/s
 - C. C. 12.5 m/s
 - D. D. 15.0 m/s
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- 25.** A ball collides with a second ball at rest. After the collision, the first ball comes to rest and the second ball moves off at the speed of the first ball. In this condition (1)
- A. A. both momentum and kinetic energy is conserved
 - B. B. only total kinetic energy is not conserved.
 - C. C. total momentum is conserved but total kinetic energy is not conserved.
 - D. D. total momentum is not conserved but total kinetic energy is conserved.
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- 26.** A man drags a trunk up the loading ramp of a truck. The ramp makes an angle of 20° with the horizontal. The man pulls with a force F that makes an angle of 30° with the ramp. The component of the force parallel to the ramp is 150 N. The mass of the trunk is 25 kg, and the coefficient of kinetic friction between the trunk and the ramp is 0.20. What is the magnitude of the acceleration of the trunk along the ramp? (1)
- A. A. 0.50 m/s^2
 - B. B. 1.20 m/s^2
 - C. C. 0.72 m/s^2
 - D. D. 0 m/s^2
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- 27.** A pot of water is placed on a stove. As the water heats up, hot water rises to the top while cooler water moves downward, forming a circulating motion. Which heat transfer mechanism is occurring in the water? (1)
- A. A. Conduction
 - B. B. Convection
 - C. C. Radiation
 - D. D. Reflection
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- 28.** Two containers hold water at the same temperature. Container X holds 1 kg of water, while container Y holds 5 kg of water. Which statement is correct? (1)
- A. A. Container Y has a larger heat capacity but the same specific heat capacity
 - B. B. Container X has a larger specific heat capacity
 - C. C. Container Y has a smaller heat capacity
 - D. D. Container X and Y have the same heat capacity
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- 29.** A 0.30 kg lead ball is heated to 90°C and dropped into an ideal calorimeter containing 0.50 kg of water initially at 20°C . What is the final temperature of the lead ball? The specific heat capacity of lead is $128 \text{ J/(kg}\cdot\text{K)}$ and the specific heat of water is $4186 \text{ J/(kg}\cdot\text{K)}$. (1)
- A. A. 84.8°C
 - B. B. 20.8°C
 - C. C. 27.8°C
 - D. D. 21.3°C

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- 30.** Which of the following is NOT correct regarding equipotential surface? (1)
- A. A. The potential difference between any two points on the surface is zero.
 - B. B. No work is done in moving the charge along the surface.
 - C. C. The electric field is always perpendicular.
 - D. D. Equipotential surface is always spherical.
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- 31.** Which of the following describes what occurs in the fission process? (1)
- A. A. A heavy nucleus is fragmented into lighter ones.
 - B. B. A neutron is split into a neutron and proton.
 - C. C. Two light nuclei are combined into a heavier one.
 - D. D. A particle and anti-particle appear in an area of high energy density.
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- 32.** Which of the following variables can be affected if a dielectric, such as a sheet of insulating plastic material, is inserted into a parallel plate capacitor (charged and disconnected from the battery) when compared to a capacitor without the material? i. Capacitance ii. Electric Field between the plates iii. Electric Potential (1)
- A. A. I and III only
 - B. B. II only
 - C. C. I and II only
 - D. D. I, II, and III
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- 33.** Which of the following statements best illustrates how physics is integral to astronomy? (1)
- A. A. Astronomy is an observational science that does not rely on physical principles because it only describes the positions of celestial objects.
 - B. B. Physics explains the motions, energy transformations, and interactions of celestial bodies, allowing predictions about stellar evolution, planetary orbits, and cosmic phenomena.
 - C. C. Physics is irrelevant in astronomy since the universe cannot be experimented upon in a laboratory.
 - D. D. Astronomy and physics are unrelated; celestial phenomena are explained entirely by mathematical models without physical laws.
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- 34.** The moment of inertia of a rigid body depends upon (1)
- A. A. linear velocity of the rotating body.
 - B. B. angular velocity of the body.
 - C. C. angular acceleration of the body.
 - D. D. distribution of mass from the axis.
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- 35.** Which of the following statements best explains why CT scans can detect abnormalities that may be hidden in conventional X-ray images? (1)
- A. A. CT scans use more penetrating X-rays, which completely pass through dense tissues, revealing hidden structures.
 - B. B. CT scans produce images that represent the cumulative absorption along a single beam path, making overlapping structures less visible.
 - C. C. CT scans reconstruct multiple angular projections into cross-sectional slices, removing the overlap of tissues along the X-ray path, and enhancing visualization of subtle differences in density.
 - D. D. CT scans rely on the radioactive decay of injected isotopes to differentiate tissues, whereas X-rays do not.
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- 36.** A projectile is fired from the surface of the Earth with a speed of 200 m/s at an angle of 45° above the horizontal. If the ground is level, what is the maximum height reached by the projectile? (1)
- A. A. 1000 m
 - B. B. 2000 m
 - C. C. 3500 m
 - D. D. 4000 m

37. A uniform rod of length $L = 2\text{ m}$ and mass $M = 5\text{ kg}$ is pivoted at one end and held horizontally. A force $F = 20\text{ N}$ is applied perpendicular to the rod at its free end. What is the angular acceleration of the rod? Use $I = ML^2/3$

A. A. 2.4 rad/s^2 B. B. 6.0 rad/s^2
C. C. 10.0 rad/s^2 D. D. 5.0 rad/s^2

38. According to Kepler, planets move in

A. A. circular orbits around the sun.
B. B. straight lines with constant velocity.
C. C. elliptical orbits around the Sun with Sun at exact centre.
D. D. elliptical orbits around the Sun with Sun at one of its foci.

39. The mass and radius of the moon are less than the mass and radius of the earth. If an object is taken from the earth to the moon, the acceleration due to gravity it experiences on the surface of the moon is less than that which it experiences on the surface of the earth. Which one of the following statements explains this phenomenon?

A. A. The object is far from the earth.
B. B. There is no atmosphere on the moon.
C. C. The radius of the moon is less than that of the earth.
D. D. The mass of the moon is less than that of the earth.

40. The density of a certain gas at 300 K is 1.8 kg/m^3 . If the molar mass of the gas is 0.029 kg/mol , what is the pressure of the gas?

A. A. $1.55 \times 10^5\text{ Pa}$ B. B. $1.20 \times 10^5\text{ Pa}$
C. C. $2.00 \times 10^5\text{ Pa}$ D. D. $0.90 \times 10^5\text{ Pa}$

41. A container is filled with oil and fitted on both ends with pistons. The area of the left piston is 10 cm^2 ; that of the right piston $10,000\text{ cm}^2$. What force must be exerted on the left piston to keep the $10,000\text{ N}$ car on the right at the same height?

A. A. 10 N B. B. 100 N
C. C. 1000 N D. D. $10,000\text{ N}$

42. A solid metal object of volume $2.5 \times 10^{-3}\text{ m}^3$ is completely submerged in water. The density of water is 1000 kg/m^3 . What is the buoyant force acting on the object?

A. A. 12.25 N B. B. 25 N
C. C. 49 N D. D. 98 N

43. Which of the following statements best distinguishes steady (laminar) flow from turbulent flow?

A. A. In steady flow, the velocity of fluid particles changes randomly with time, while in turbulent flow it remains constant.
B. B. In steady flow, the fluid velocity is always zero, while turbulent flow occurs only when fluid is stationary.
C. C. In steady flow, pressure variations are absent, whereas turbulent flow occurs only in compressible fluids.
D. D. In steady flow, fluid particles move in smooth parallel layers with little mixing, whereas turbulent flow is characterized by irregular motion and strong mixing between layers.

44. A long straight conductor carries a steady electric current of 10 A . During a laboratory experiment, a magnetic field probe measures the magnetic field strength around the wire. If the magnetic field at a certain point is measured to be $4 \times 10^{-5}\text{ T}$, what is the distance of the probe from the wire?

A. A. 2 cm B. B. 5 cm
C. C. 8 cm D. D. 10 cm

- 45.** A square coil of wire with a side length of 10 cm is looped around 10 times. The coil sits in an increasing magnetic field. The magnetic field increases linearly from 1T to 2T within 5 seconds. If the same loop of wire has a resistance of 2Ω , what is the induced current in the loop? (1)
- A. A. 100 A
B. B. 10 A
C. C. 1 A
D. D. 0.01 A
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- 46.** Which of the following statements best explains the I–V characteristic of the diode in this forward-bias region? (1)
- A. A. The current increases linearly with voltage from the moment the voltage is applied.
B. B. The current increase slowly at first and then rises rapidly after the threshold voltage is reached.
C. C. The current decreases as the voltage increases.
D. D. The current remains zero for all applied voltages.
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- 47.** Which statement best explains the working principle of a full-wave rectifier? (1)
- A. A. It allows only the positive half-cycles to pass while blocking the negative half-cycles.
B. B. It allows current to flow during both half-cycles by reversing the direction of the negative half-cycles, producing a continuous pulsating DC output.
C. C. It eliminates all voltage fluctuations and produces a constant DC voltage.
D. D. It produces alternating voltage in the same shape as the input AC.
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- 48.** Which of the following conditions are needed to properly bias an n-p-n transistor amplifier? (1)
- A. A. Forward bias the base/emitter junction and reverse bias the base/collector junction.
B. B. Forward bias the collector/base junction and reverse bias the emitter/base junction.
C. C. Apply a positive voltage on the n-type material and a negative voltage on the p-type material.
D. D. Apply a large voltage on the base.

— END OF EXAMINATION —